Drilling fluid maintains **wellbore stability**.

Key Points:

- **Prevent caving and washout:** Mud pressure supports the hole walls.
- **Smooth gauge hole:** Ensures proper hole diameter without deformation.
- **Formation consolidation:**
  - **Old formations:** Hard, consolidated → stable.
  - **Young formations:** Unconsolidated → require careful mud design.
- **Pressure and temperature with depth:**
  - As depth increases, **pressure and temperature rise**.
  - Overburden stress increases, which affects wellbore stability.

Proper mud design prevents **hole collapse, washouts, or irregular hole shapes**.

Drilling Fluid Preparation

Drilling fluid (mud) consists of **two main parts**:

(A) Continuous Phase (Base Fluid)

This is the main liquid part of mud. It can be:

- **Water** → Water-Based Mud (WBM)
- **Oil** → Oil-Based Mud (OBM)
- **Synthetic oil** → Synthetic-Based Mud (SBM)

The continuous phase controls the overall behavior of the mud.

(B) Solid Additives (Powder Form)

These are added to modify properties of the base fluid.

Main purposes of additives:

	Purpose
Weighting materials	Increase mud density (mud weight)
Shale stabilizers	Prevent shale swelling and instability
Viscosifiers	Increase viscosity and carrying capacity
Filtration control materials	Reduce fluid loss into formation
Rheology control materials	Adjust flow behavior (gel strength, plastic viscosity)
Alkalinity & pH control materials	Maintain proper chemical balance
Lost circulation materials (LCM)	Control mud losses
Lubricants	Reduce torque and drag

1. Weighting Materials (Densifiers)

Purpose:

To increase mud density (mud weight)

To control formation pressure

Also called **densifiers**

Types:

(1) Barite ($BaSO_4$)

✔ Specific Gravity (SG): **4.2 – 4.4/4.5**

✔ Most commonly used weighting material
✔ Cost effective

Limitations:

reservoir damaging behaviour

Not acid soluble

cause pore plugging (decrease porosity)

Can reduce formation permeability

(2) Iron Ore Minerals

These include:

Hematite

Magnetite

Siderite

✔ Specific Gravity: **around 5.0 – 5.2**
✔ Used for high-density mud
✔ Can prepare mud up to **22 ppg**

✔ used when very high mud weight is required.

(3) Calcium Carbonate ($CaCO_3$)

✔ Specific Gravity: **2.6 – 2.8**
✔ Used for mud up to about **12 ppg**

Advantages:

Reservoir friendly

Acid soluble

HCl can dissolve it — this helps remove pore plugging and reduce formation damage.

(4) Micromax & Microdense

These are high-performance weighting materials.

✔ Specific Gravity: **4.7 – 4.9**
✔ Alternative to barite
✔ Finer particle size
✔ Better suspension properties
✔ Reduce sagging problem

SHALE STABILIZING MATERIALS

1. What is Shale?

Shale is a **clastic sedimentary rock**.

It is mainly composed of:

- Mud
- Silt
- Clay minerals

Common Clay Minerals in Shale:

- **Kaolinite**
- **Illite**
- **Smectite**
- **Chlorite**

These clay minerals are responsible for shale reactivity and swelling behavior.

2. Why is Shale Stabilization Required? (UP)

Approximately **40% of (stratigraphic column) sedimentary rocks** are shale.

Around **70% of wellbore problems** occur due to shale instability.

Therefore, shale stabilization is necessary to:

- Prevent wellbore collapse
- Reduce swelling
- Avoid hole enlargement
- Prevent stuck pipe problems

3. Shale Structure (Plate-Type Structure)

Shale (especially clay minerals like smectite) has a **plate-like structure**.

Important Characteristics:

- Clay particles are arranged in thin plates.
- Plate surfaces are **negatively charged**.
- The distance between plates is called **Basal Spacing**.
- Basal spacing is measured in **Angstrom (Å)**.
  - 1 Å = 10^{-10} meters

4. Swelling Mechanism in Water-Based Mud (WBM)

When Water-Based Mud is used:

- Water contains **H^+ ions**.
- These positive ions are attracted to the **negatively charged clay surfaces**.
- Water molecules enter between the plates (basal spacing).
- Basal spacing increases.
- This causes **hydration and swelling of shale**.

This process is called **Shale Hydration Swelling**.

5. Shale Swelling Consequences

If swelling is not controlled:

- Wellbore instability
- Tight hole
- Stuck pipe
- Caving (shale pieces falling into wellbore)

6. Shale Swelling Reduction Methods

To control swelling, drilling fluid is modified.

(A) Salts

Salts reduce swelling by blocking or stabilizing basal spacing.

Common salts:

- $NaCl$
- KCl
- $MgCl_2$

Mechanism:

- Salt ions replace exchangeable ions in clay.
- Reduce water entry.
- Decrease swelling.

Salt ions replace exchangeable ions in clay - Explanation (UP)

- Salts (like $NaCl$) dissolve into Na^+ ions in drilling fluid.
- Na^+ ions **replace the clay's exchangeable cations** (cation exchange).
- Large salt particles **do not physically enter** the basal spacing.
- Na^+ reduces the negative charge on clay plates, **reducing electrostatic repulsion**.
- Less water enters the basal spacing → **swelling is minimized**, but a small amount of water may still enter.
- Result: shale becomes **stable** without significant expansion.

Electrostatic repulsion is the force that pushes negatively charged clay plates away from each other.

Molecular Weight Effect (UP)

Both **H^+ and Na^+** are positive ions, but their **molecular size/weight** affects swelling.

H^+ ions are small → less electrostatic repulsion reduction → attract more water → **more swelling**.

Na^+ ions are larger → reduce repulsion more strongly and form a tighter hydration layer → **less water enters → swelling is controlled**.

(B) Polymers

Polymers:

- Form a **gel-type structure**
- Coat the clay surface
- Reduce water penetration
- Stabilize the plate structure

They create a protective layer over the shale.

(C) Nanoparticles

Nanoparticles:

- Enter and block basal spacing
- Prevent water penetration
- Stick inside micro-pores

Common nanoparticles:

- ZnO (Zinc Oxide)
- TiO_2 (Titanium Oxide)
- CuO (Copper Oxide)
- SiO_2 (Silica Oxide)
- Graphene Oxide
- Multi-walled Carbon Nanotubes (MWCNTs)

Mostly used:

- Silica Oxide
- Graphene Oxide

7. Shale Testing Methods

(1) XRD (X-Ray Diffraction)

Full name: **X-Ray Diffraction**

Purpose:

- Determines mineralogy of shale.
- Shows amount of each mineral in rock.
- Identifies clay type.

(2) CEC Test (Cation Exchange Capacity)

Also called:

- Methylene Blue Test (MBT)
- Titration Method

Meaning of CEC: (UP)

CEC measures the ability of clay to exchange cations.

- High CEC → High swelling potential
- Low CEC → Low swelling

Low CEC clay → fewer negatively charged regions on the clay surface → fewer sites for ions to attach → Na^+ effect dominates → less swelling.

High CEC clay → many negatively charged regions → more sites for ions → H^+ can attach easily → more water enters → more swelling.

CEC Procedure

- Take 1 gram fine shale powder → (10 meq/100g → mean **1 g of fine shale powder is mixed into 100 g of drilling fluid containing 10 mL equivalent of whatever reagent** (e.g., salt, acid, or indicator).
- Add:

- Distilled water
- Shale powder
- Sulfuric acid
- Methylene Blue (MBT)

Shake the flask.

Put a drop on filter paper.

When a light green/green ring appears around the blue drop → Stop.

Amount of MBT used determines CEC value.

Higher MBT consumption = Higher CEC = Higher swelling tendency.

8. Linear Dynamic Swell Meter (LDSM) Test

Purpose:
To measure shale swelling percentage over time.

Sample Preparation

- Collect shale (drill cuttings).
- Dry in oven:
    - 105–110°C
    - For 3 hours
- Grind using mortar grinder.
- Sieve using shaker:
    - Less than 0.212 mm size selected.
- Take 15 grams.
- Compress using hydraulic compactor:
    - 8000 psi
    - 30 minutes
      → Forms shale pellet.

Testing Procedure

- Place pellet inside LDSM.
- Add 175 ml drilling fluid.
- Transducer measures swelling.
- Computer records data.
- Graph plotted:
    - Y-axis: Swelling %
    - X-axis: Time

In the **LDSM test**: **(UP)**

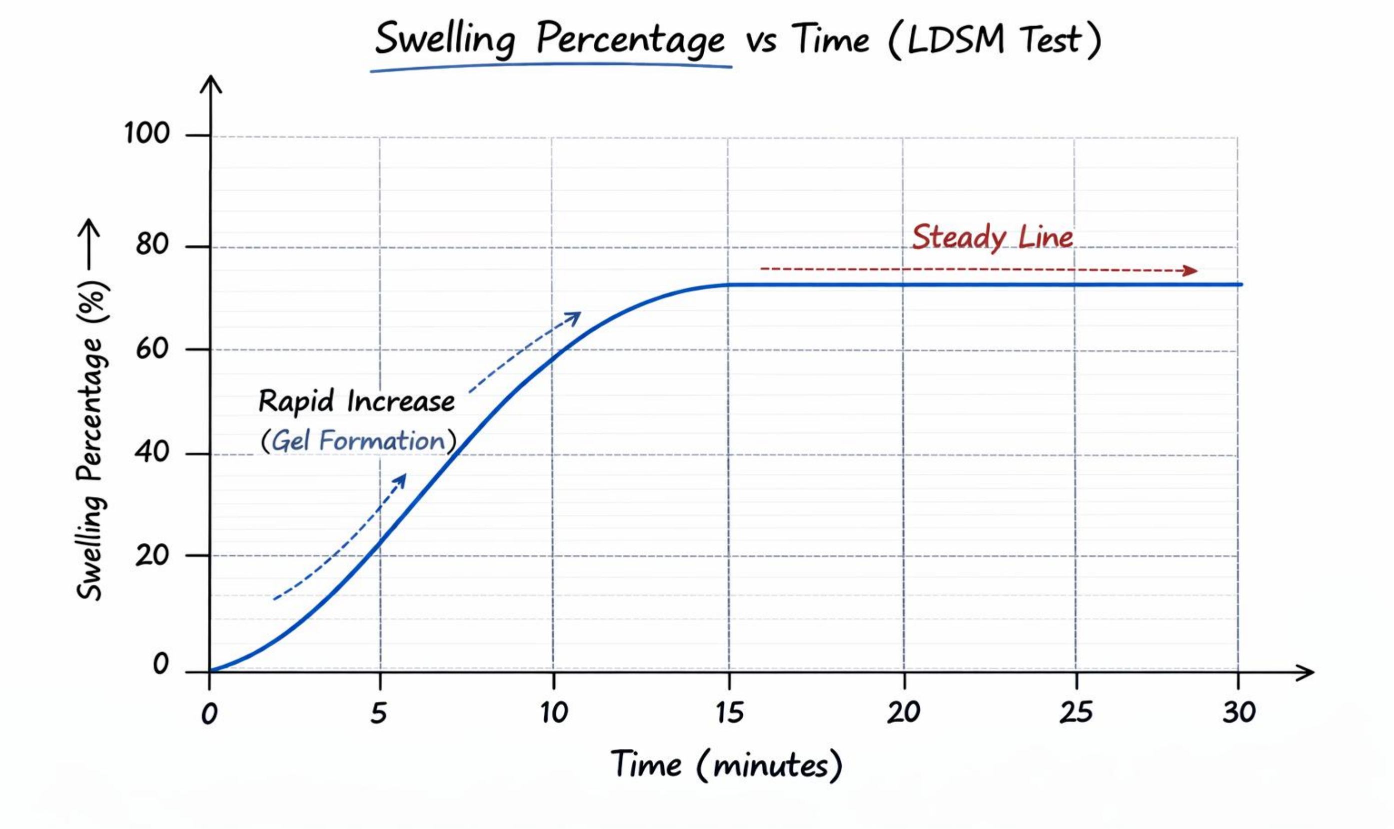
- The shale pellet is **confined from all sides except the top**, where the **transducer** is placed.
- When the pellet **swells**, even **small changes in height** push the transducer upward.
- The transducer detects this movement and converts it into a **linear swelling curve** over time.
- This is why the LDSM measures **linear swelling in one direction only** (upward).

Graph Behavior

Initially swelling increases (curve rises like parabola).

Then reaches steady line.

When steady line appears → swelling stabilized → stop test.



Viscosifiers

Purpose:

Increase the **viscosity** of the drilling fluid.

Help **suspend drill cuttings and weighting materials**.

Keep solids **dispersed** in the fluid, preventing settling.

Common Viscosifiers:

Bentonite Clays

Most commonly used viscosifier.

Sodium bentonite → high swelling, good viscosity.

Calcium bentonite → lower swelling, lower viscosity.

Performance:

Good near the surface.

At depth/high temperature → performance may degrade.

XRD test shows the structural difference between sodium and calcium bentonite.

Polymers

PHPA (Partially Hydrolyzed Polyacrylamide) → synthetic polymer, stable at depth, controls fluid loss.

Sodium alginate → natural biopolymer; must **form a gel before use**, not added directly.

Xanthan gum → natural biopolymer; performs better at depth and high temperature compared to bentonite.

CMC (Carboxymethyl cellulose) → semi-synthetic polymer, helps control viscosity and fluid loss.

PAC (Polyanionic cellulose) → synthetic polymer, water-soluble, enhances viscosity.

Key Concepts: (UP)

Settling Effects:

If drill cuttings settle:

Viscosity decreases

Density reduces

Pressure reduces

To prevent this, viscosifiers maintain **suspension**.

Gel Strength:

The property of drilling fluid that allows it to **hold cuttings in suspension** even when circulation stops.

Stronger gel → better suspension, but very high gel strength → harder to pump.

Rheology Control Materials

Rheology Definition:
Rheology is the study of flow and deformation of fluids. In drilling fluids, it helps determine the **nature and properties of the mud**, such as how it flows, suspends cuttings, and responds under stress.

Properties Measured in Rheology:

Plastic Viscosity (PV) – flow resistance due to internal friction of solids and liquid layers combined.

Yield Point (YP) – initial stress required to start fluid movement.

Gel Strength – ability of fluid to suspend cuttings when circulation stops.

Apparent Viscosity (AV) – effective viscosity at a specific shear rate.

Flow Behavior / Transport Index – indicates cuttings transport efficiency.

Measuring Instruments:

Rheometer / Marsh Funnel / Viscometer

Typical **speeds**: 600, 300, 200, 100, 6, 3 RPM

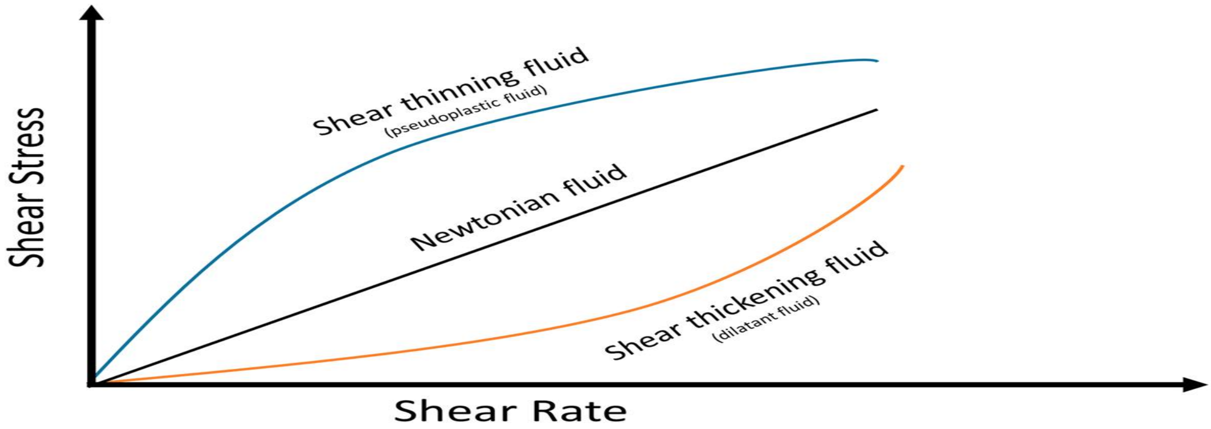
Rheometer Types: Standard, High Temperature (HT), High Pressure High Temperature (HPHT)

Shear Stress vs. Shear Rate Graph:

Shows **shear thinning /** pseudoplastic (viscosity decreases with increasing shear rate) or **shear thickening** / dilatant (viscosity increases with increasing shear rate).

Degree of shear thinning reflects how fluid behavior changes under stress.

Degree of shear thickening reflects how much the fluid's viscosity increases as shear stress increases.



Term	Definition	Unit
Shear Rate ($\dot{\gamma}$)	Rate at which adjacent fluid layers move relative to each other	s^{-1}
Shear Stress (τ)	Force per unit area applied parallel to the fluid layers, causing flow	Pa or lb/100 ft^2

Formulas:

- **Plastic Viscosity (PV):** PV = θ600 – θ300
- **Yield Point (YP):** YP = θ300 – PV
- **Apparent Viscosity (AV):** AV = θ600 ÷ 2
- **Transport Index (TI):** TI = YP ÷ PV

Plastic Viscosity (PV):

- PV is the **combined viscosity** of solid particle interactions and water layer viscosity.
- **API Standard (Water-Based Mud):** RP 13B-1, recommended 8–35 cP
  - PV < 8 → High cutting accumulation → increased pressure
  - PV > 35 → Flow problems
  - Target: **<5% remaining cuttings** in mud

Low Plastic Viscosity (PV < 8 cP):

- If PV is too low, the drilling fluid **cannot suspend cuttings properly**.
- This leads to **cuttings accumulation in the wellbore**.
- As cuttings build up, the **resistance to flow increases**, so the **pressure inside the wellbore rises.**

High Plastic Viscosity (PV > 35 cP):

- If PV is too high, the fluid becomes **too thick to flow easily**.
- This causes **flow problems**, meaning it is **difficult to circulate the mud** and clean the wellbore.

Yield Point (YP):

- Unit: **lb/100 ft^2**
- API Standard: same as PV
- Min: 5 lb/100 ft^2; Max: ≤3×PV
- Graph Explanation:
  - **Water line** starts from zero because water has no initial solids to push.
  - **Mud line** does not start at zero due to **initial stress caused by solid particles in the mud**, which requires extra force to initiate flow.